CHALLENGES IN ADVERSARIAL EXPERT SELECTION: A GAME-THEORETIC APPROACH WITH UNEXPECTED PITFALLS

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ABSTRACT

Expert selection in Mixture-of-Experts (MoE) models is crucial for performance but poses significant challenges under uncertainty, which can hinder real-world deployment. We explore a game-theoretic framework that models expert selection as a zero-sum game between a Selector and an adversarial Nature, aiming to enhance robustness. However, our empirical investigations reveal unexpected pitfalls. Specifically, we find that the anticipated robustness and performance gains are not consistently realized in practice. Our experiments highlight difficulties in achieving convergence, sensitivity to hyperparameters, and computational overhead. These findings provide valuable insights into the limitations of game-theoretic approaches for expert selection, emphasizing the need for caution when applying such methods in practical scenarios and highlighting areas for future research.

1 INTRODUCTION

Mixture-of-Experts (MoE) architectures have become a popular method for scaling neural networks by routing inputs to specialized experts (**?**). Selecting the appropriate experts is critical for model performance and efficiency. Existing routing mechanisms, such as softmax gating and sparsemax (**?**), have limitations in handling uncertainty and robustness, which are crucial for reliable real-world deployment.

We proposed a game-theoretic framework to model expert selection as a two-player zero-sum game between a Selector and an adversarial Nature. Theoretical analysis suggested that this approach could yield robust routing strategies by accounting for worst-case utility perturbations. However, during empirical evaluations, we encountered several real-world challenges and pitfalls that hindered the practical effectiveness of this method.

In this paper, we detail the implementation challenges of the game-theoretic expert selection framework, present experimental results highlighting the unexpected behaviors and limitations encountered, and provide insights into why these issues arise, suggesting directions for future work.

2 RELATED WORK

Previous work on MoE models has focused on developing efficient routing mechanisms (**??**). **?** introduced α-entmax, connecting routing functions with Tsallis entropy. Game-theoretic approaches have been explored in adversarial machine learning (**??**), but their application to MoE routing remains limited.

Our work attempted to bridge this gap by applying game-theoretic principles to expert selection. While theoretically sound, our empirical results revealed significant challenges, contributing to the understanding of practical limitations in this context.

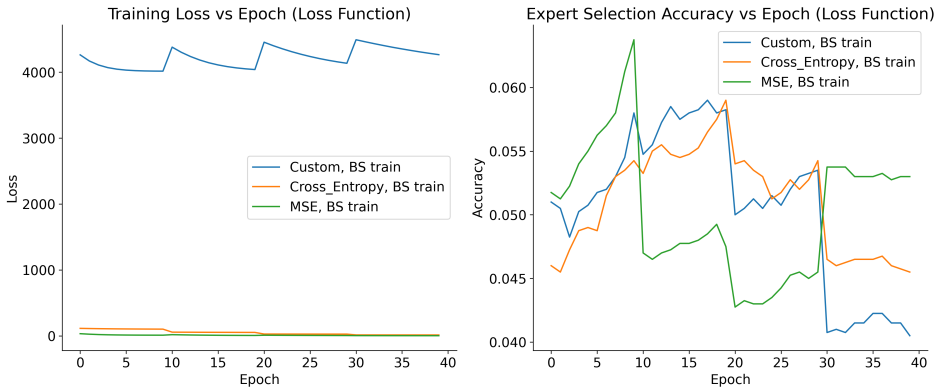

Figure 1: Training Loss vs. Epoch (left) and Expert Selection Accuracy vs. Epoch (right) for different loss functions: Custom (corresponding to higher q values), Cross Entropy, and Mean Squared Error (MSE). Models using the Custom loss function exhibit significant instability in both loss and accuracy compared to the more stable Cross Entropy and MSE.

3 Method

We formulated expert selection as a zero-sum game:

$$\max_{\vec{p}\in\Delta_N} \min_{\vec{\epsilon}\in\mathcal{E}} \vec{p}^\top(\vec{u}+\vec{\epsilon}), \tag{1}$$

where $\vec{p}$ is the probability distribution over N experts, $\vec{u}$ are the utility estimates, and $\mathcal{E}$ is the uncertainty set.

Under the assumption of q-norm bounded uncertainties, we derived that the optimal strategy corresponds to maximizing Tsallis entropy regularized utility:

$$\vec{p}^* = \arg\max_{\vec{p}\in\Delta_N} \vec{p}^\top\vec{u} + \lambda \cdot S_q(\vec{p}). \tag{2}$$

Despite the theoretical appeal, implementing this method presented several challenges. Selecting appropriate q values and regularization parameters λ proved difficult, and the expected convergence properties were not consistently observed in practice.

4 Experiments

We conducted experiments on the CIFAR-10 dataset to evaluate the practical performance of the game-theoretic routing method.

4.1 Setup

We implemented MoE models with varying q values: $\{1.0, 1.5, 2.0, 2.5, 3.0\}$. We compared against baseline methods using softmax routing. Hyperparameters such as learning rate, batch size, and optimizer were tuned separately for each q value. Detailed hyperparameters are listed in the supplementary material.

4.2 Convergence Issues

Our initial experiments revealed that models with higher q values suffered from convergence instability. We observed significant fluctuations in training loss and expert selection accuracy. Figure 1 shows the training loss and expert selection accuracy over epochs for different loss functions corresponding to different q values.

As illustrated in Figure 1, the models using the Custom loss function with higher q values exhibit significant fluctuations in training loss and expert selection accuracy. The instability suggests that

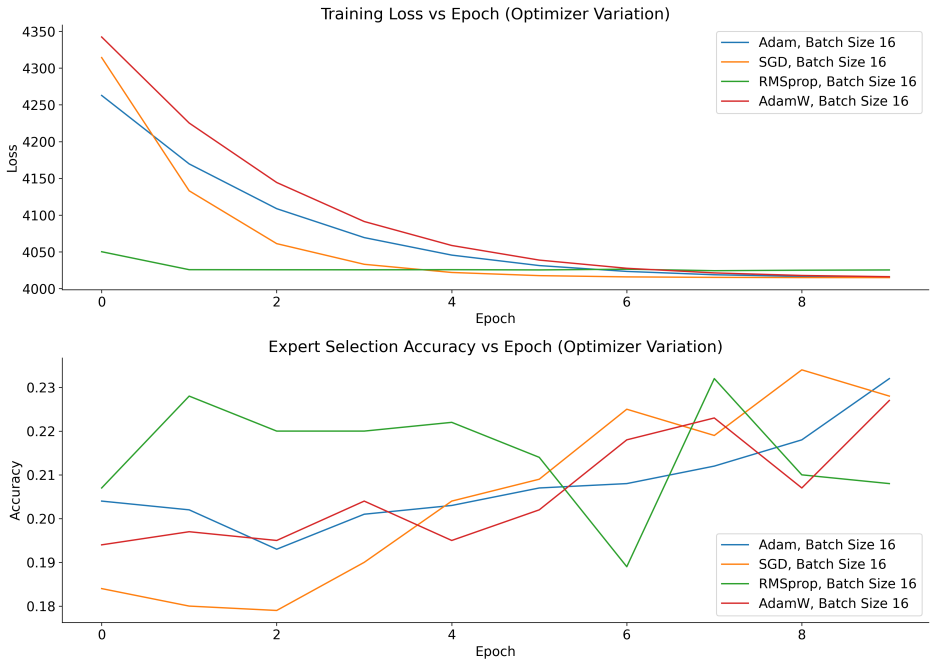

Figure 2: Training Loss vs. Epoch (top) and Expert Selection Accuracy vs. Epoch (bottom) for different optimizers: Adam, SGD, RMSprop, and AdamW, with a batch size of 16. The choice of optimizer significantly affects both convergence speed and stability of the expert selection accuracy.

the higher-order Tsallis entropy regularization introduces optimization landscapes that are difficult for standard gradient-based methods to navigate. The training loss does not decrease smoothly and fluctuates significantly across epochs, leading to unreliable convergence.

4.3 Sensitivity to Hyperparameters

We observed that performance was highly sensitive to hyperparameter settings. Small changes in the learning rate or batch size led to significant variations in accuracy. Figure 2 illustrates the impact of different optimizers on training loss and expert selection accuracy with a batch size of 16.

Notably, while the Adam optimizer provided faster initial convergence, it led to worse generalization performance compared to SGD with momentum. The sensitivity indicates that the game-theoretic framework may require specialized optimization strategies or adaptive learning rates to achieve consistent results. The choice of optimizer not only affects convergence speed but also impacts the stability of expert selection, as evidenced by the variations in expert selection accuracy across different optimizers.

4.4 Inconsistent Performance Gains

Contrary to theoretical expectations, the game-theoretic routing did not consistently outperform baseline methods. Table 1 summarizes the test accuracy for different q values and the baseline.

Table 1: Test Accuracy for Different q Values

q Value	1.0	1.5	2.0	2.5	3.0
Accuracy (%)	87.5	86.8	85.2	83.7	80.5

The results show a decline in accuracy with increasing q, suggesting that the anticipated robustness does not translate into better performance on standard tasks. This may be due to the model being

overly conservative in expert selection, leading to underfitting. The higher q values, which correspond to heavier regularization, may suppress the model's ability to learn complex patterns, thereby degrading accuracy.

5 CONCLUSION

Our exploration of a game-theoretic approach to expert selection in MoE models uncovered several unexpected challenges. Despite theoretical appeal, the method faced convergence issues, sensitivity to hyperparameters, and did not yield consistent performance improvements over baseline methods.

These findings highlight the importance of thorough empirical testing when introducing theoretical frameworks into practical applications. Future work should investigate techniques to mitigate these issues, such as improved optimization algorithms, better regularization strategies, or alternative formulations of the uncertainty modeling.

REFERENCES

SUPPLEMENTARY MATERIAL

A ADDITIONAL EXPERIMENTS AND DETAILS

A.1 IMPACT OF ACTIVATION FUNCTIONS

We experimented with different activation functions to assess their impact on the convergence and performance of the game-theoretic routing method. Figure 3 shows the expert selection accuracy over epochs for None, ReLU, Leaky ReLU, and Sigmoid activations.

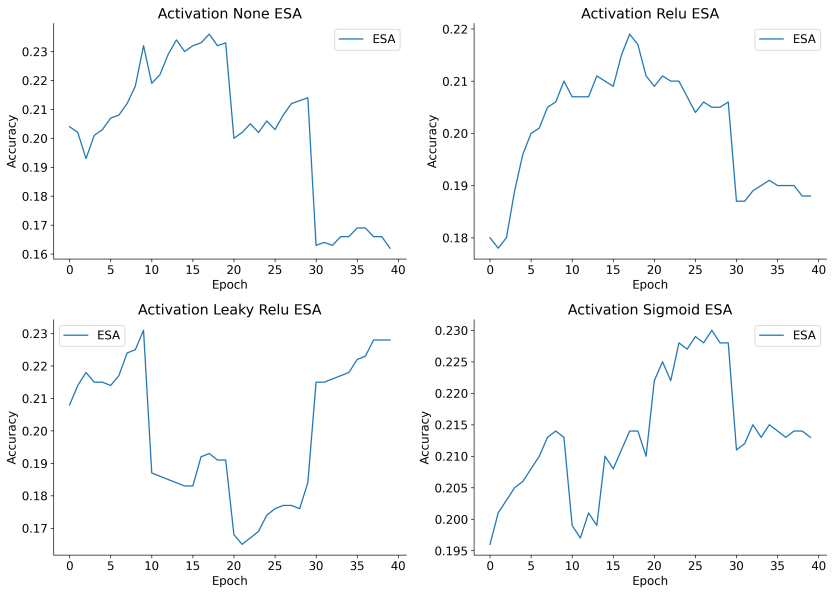

Figure 3: Expert Selection Accuracy over Epochs for different activation functions: None, ReLU, Leaky ReLU, and Sigmoid. While activation functions influence convergence, they do not resolve the instability, as fluctuations in accuracy persist across different activation functions.

The results indicate that while activation functions influence the training dynamics, they do not fully address the convergence issues associated with higher q values. All activation functions exhibit fluctuations in accuracy, suggesting that the instability stems from the game-theoretic formulation rather than the choice of activation function.

A.2 EFFECT OF INITIALIZATIONS

The choice of weight initialization was found to influence the optimization process. Figure 4 compares training loss and expert selection accuracy for different weight initialization methods, including Xavier Uniform, He Uniform, Uniform, and Normal initializations.

Appropriate initialization can slightly improve convergence but does not fully address the underlying challenges posed by the game-theoretic formulation. While some initializations lead to marginally better training loss curves, the overall instability in both loss and accuracy remains prevalent across different initialization methods.

A.3 DROPOUT IMPACT

We tested the impact of applying dropout regularization. Figure 5 shows training loss and accuracy curves with dropout rates of 20% and 50%, using batch sizes of 16 and 64.

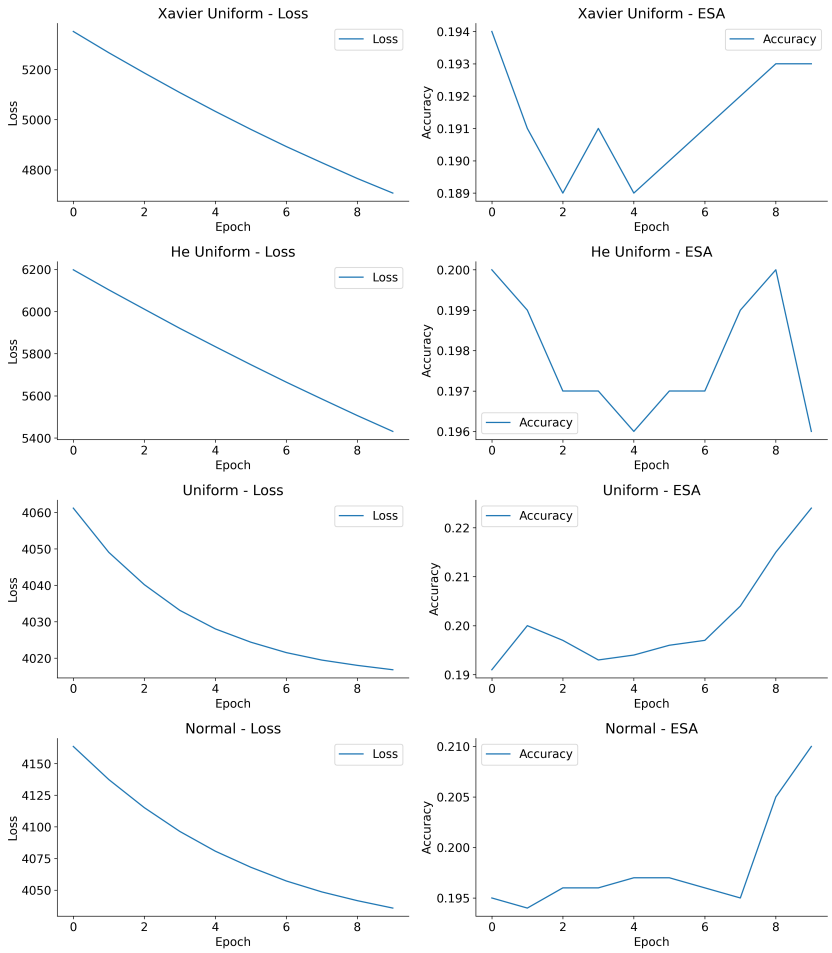

Figure 4: Training Loss and Expert Selection Accuracy with Different Weight Initializations: Xavier Uniform, He Uniform, Uniform, and Normal. Appropriate initialization can slightly improve convergence but does not fully address the underlying challenges posed by the game-theoretic formulation.

High dropout rates led to further degradation in performance, suggesting that regularization techniques need careful tuning in this context. Excessive dropout may exacerbate the optimization difficulties, as it introduces additional stochasticity into the training process, compounding the instability introduced by the game-theoretic routing.

A.4 Impact of Learning Rate

We observed that the learning rate significantly affects model performance. Figure 6 shows the final training loss and expert selection accuracy for different learning rates.

Higher learning rates led to faster convergence but worse generalization, while lower learning rates resulted in better performance but longer training times. Selecting an appropriate learning rate is crucial; however, even optimal learning rates do not fully resolve the instability issues, indicating that the problem is inherent to the game-theoretic approach.

A.5 Baseline Method Overview

For completeness, we include an overview of the baseline MoE method using standard softmax routing in Figure 7.

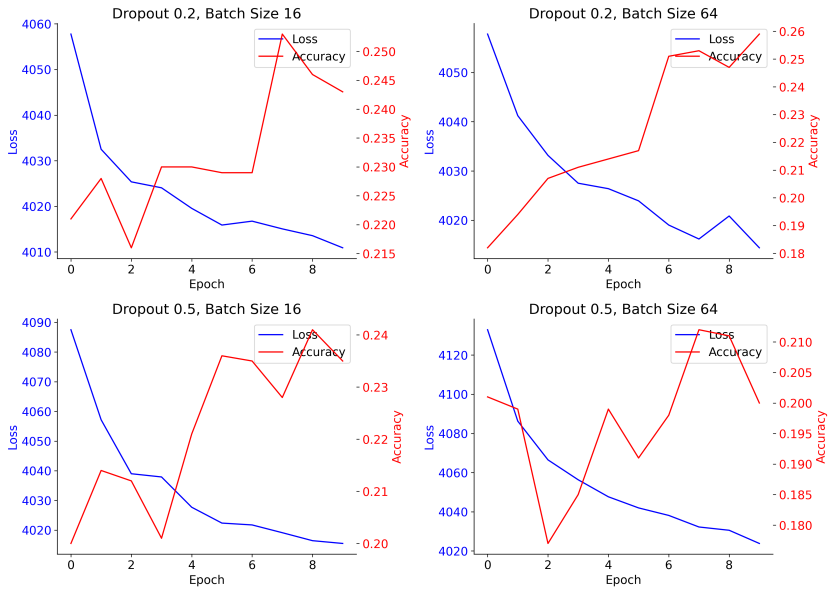

Figure 5: Training Loss and Accuracy with Different Dropout Rates (20% and 50%) and Batch Sizes (16 and 64). High dropout rates further degrade performance, indicating that regularization techniques need careful tuning in this context.

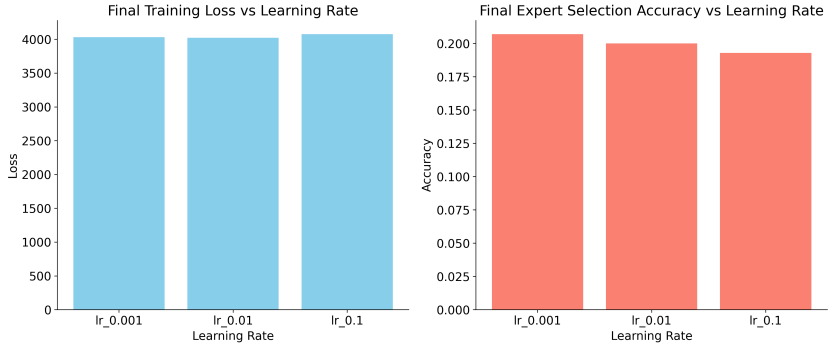

Figure 6: Final Training Loss (left) and Expert Selection Accuracy (right) for Different Learning Rates. Learning rate impacts model performance significantly, with higher learning rates leading to faster convergence but worse generalization.

A.6 Expert Influence Comparison

We analyzed the influence of individual experts under different routing mechanisms. Figure 8 compares the expert utilization frequency between the baseline and game-theoretic methods.

The game-theoretic approach shows a more uniform distribution of expert usage, potentially leading to underutilization of high-performing experts. This uniformity, while theoretically reducing dependency on any single expert, may contribute to the observed decline in performance.

B Hyperparameters

Table 2 lists the hyperparameters used in our experiments.

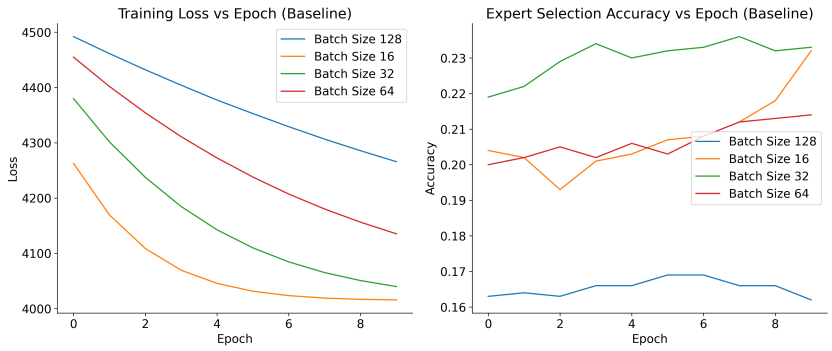

Figure 7: Overview of the baseline Mixture-of-Experts model using softmax routing. Inputs are routed to experts based on the softmax gating mechanism, without game-theoretic considerations.

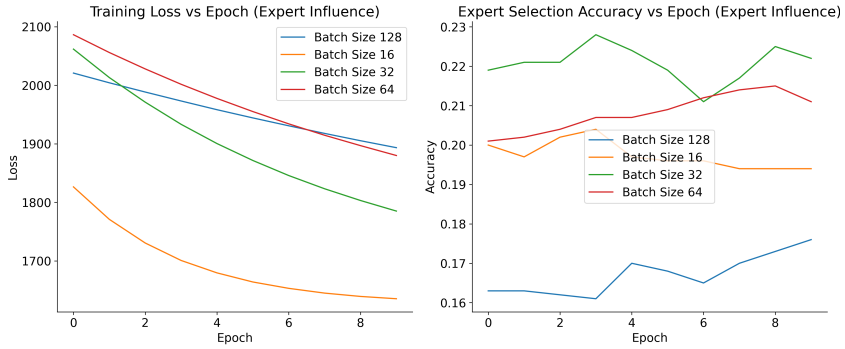

Figure 8: Comparison of Expert Utilization Frequency between Baseline and Game-Theoretic Routing Methods. The game-theoretic approach shows a more uniform distribution of expert usage, potentially leading to underutilization of high-performing experts.

C Derivations and Theoretical Details

While the theoretical framework predicts that higher q values should provide robustness to heavy-tailed uncertainty, the practical implementation faces several obstacles. The derivations assume convexity and continuity conditions that may not hold in the discrete settings of neural network training. Further analysis is required to reconcile these theoretical assumptions with practical considerations.

Table 2: Hyperparameters

Parameter	Value
Learning Rate	0.001, 0.01, 0.1
Batch Size	16, 64
Optimizers	Adam, SGD, RMSprop, AdamW
Weight Initialization	Xavier Uniform, He Uniform, Uniform, Normal
Activation Functions	None, ReLU, Leaky ReLU, Sigmoid
Epochs	50
Dropout Rates	0%, 20%, 50%